

Domain-Aware Mitotic Figure Detection for Histology Images

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Problem Formulation

10 million

Cancer deaths in 2020

Mitotic Count

Gold standard for grading tumor aggressiveness, but has inter-operator variability

Domain Bias

Features cluster by tissue/lab/scanner, not biology



Goal

Build a domain-invariant AI model for mitotic figure detection that generalizes across any tumor, scanner, lab, or species

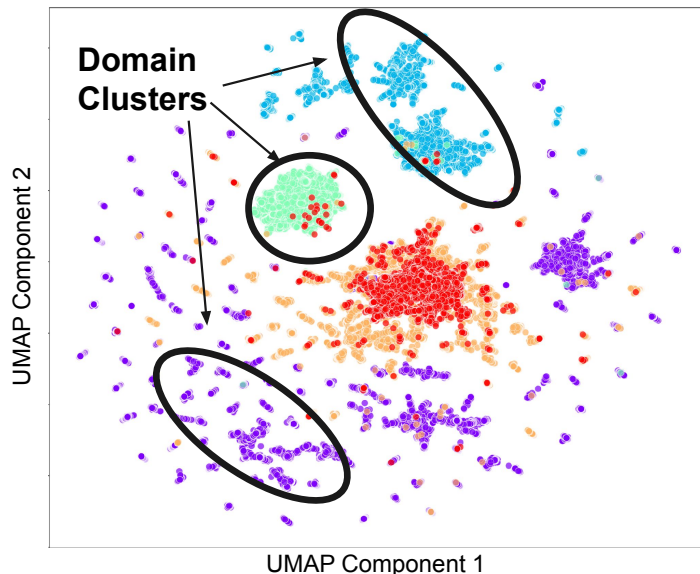
THE BARRIER

Computational Challenge

Pathologist: Interprets biological morphology

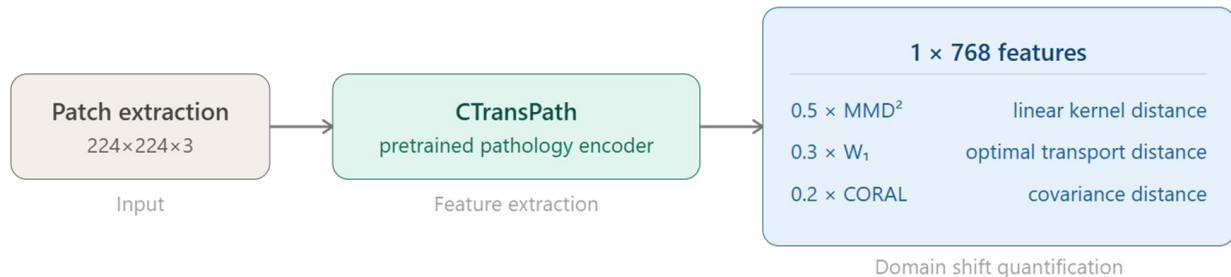
AI: Overfits to stochastic artifacts

Baseline features cluster by scanner/domain, not by mitotic status



Colors indicates different scanners

Quantifying Domain Shift



Domain Bias Score
(Higher = Greater Bias)

70–74

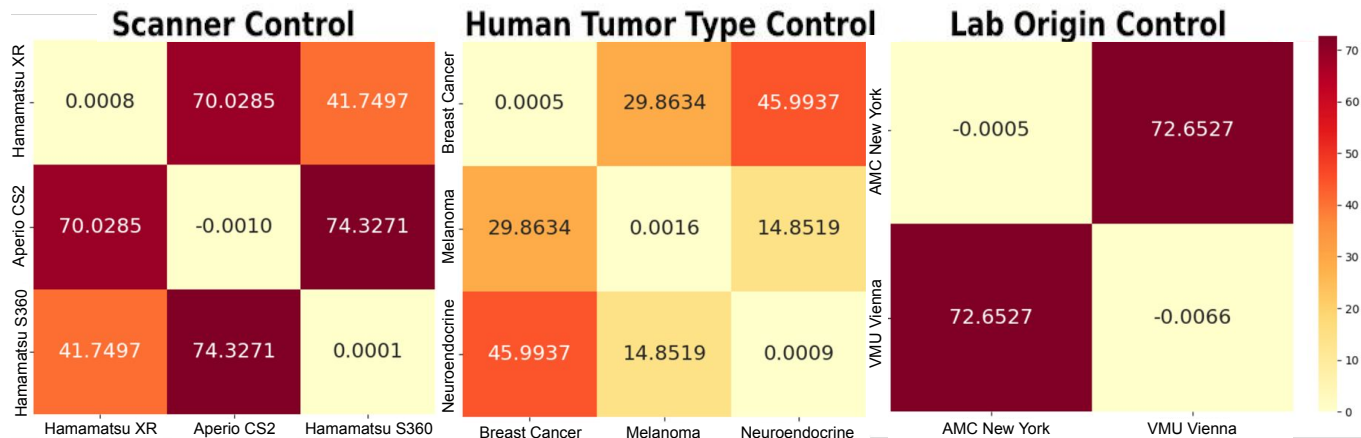
Scanner Variance

72.7

Lab Variance

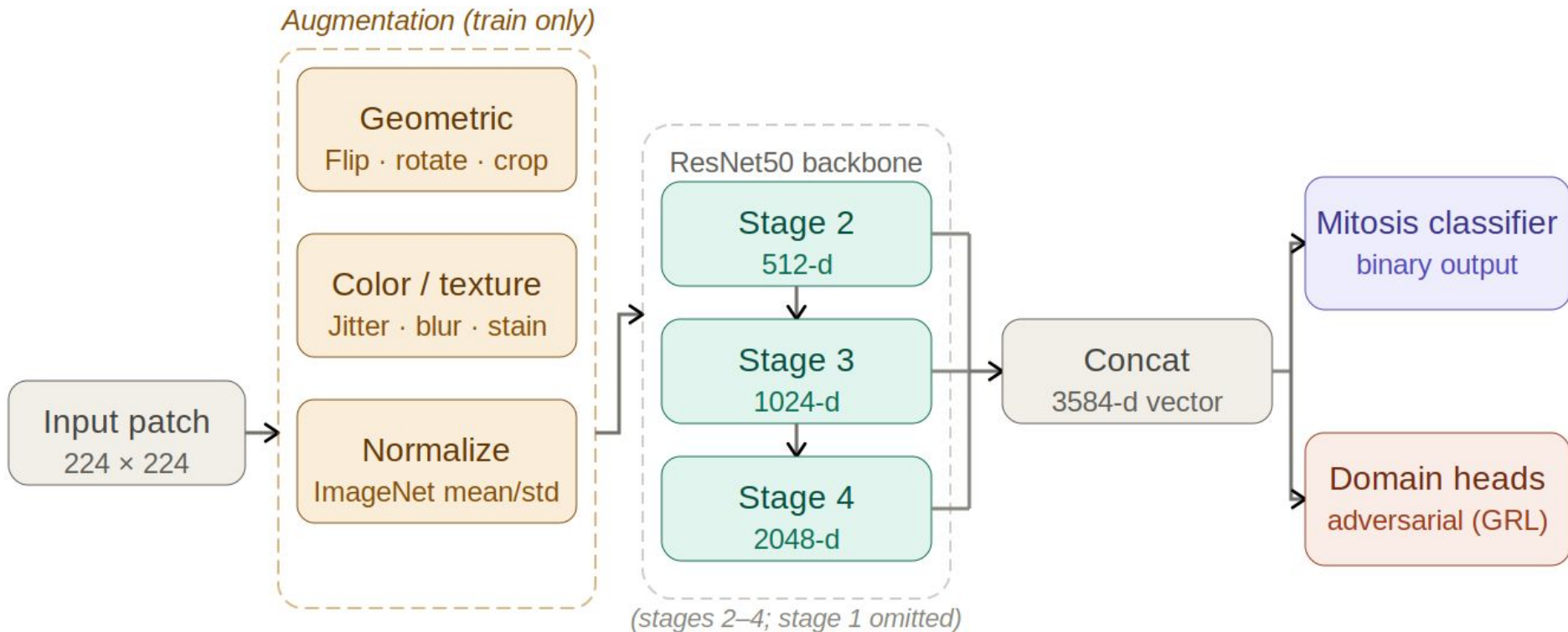
15–46

Biological Variance



Technical noise dominates over biological signal

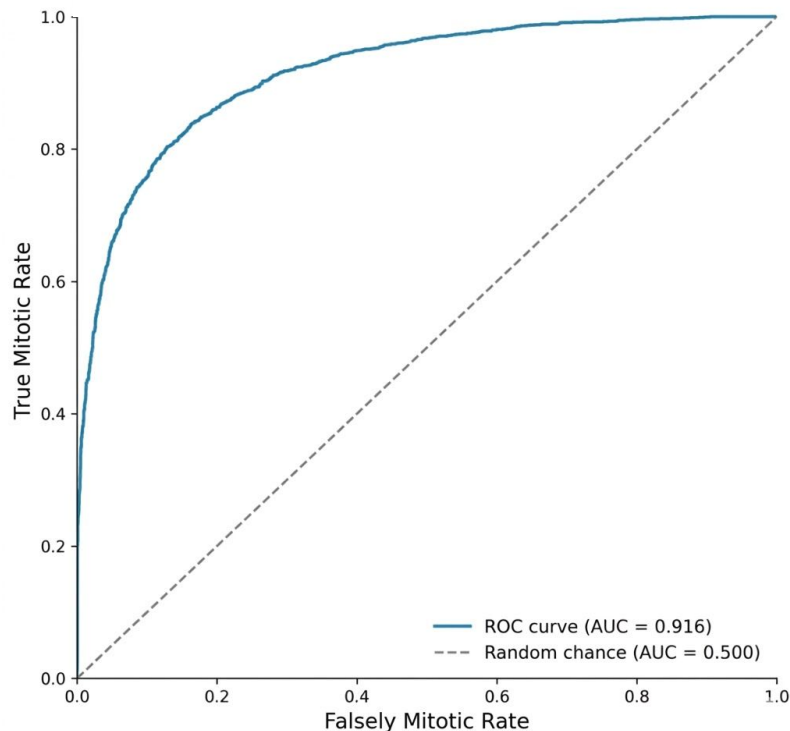
Solution: Multi-Scale Adversarial Training



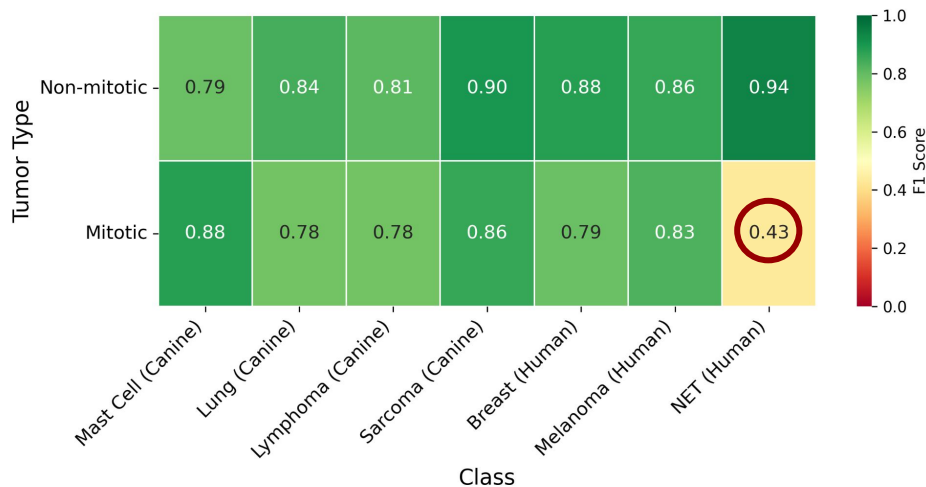
Domain shift across scanners and tissue origins is the core challenge, so we combine geometric, color, and stain augmentation with a multi-scale ResNet50 backbone and adversarial training to improve generalization.

Results

ROC Curve - Mitosis Classifier (Test Set)



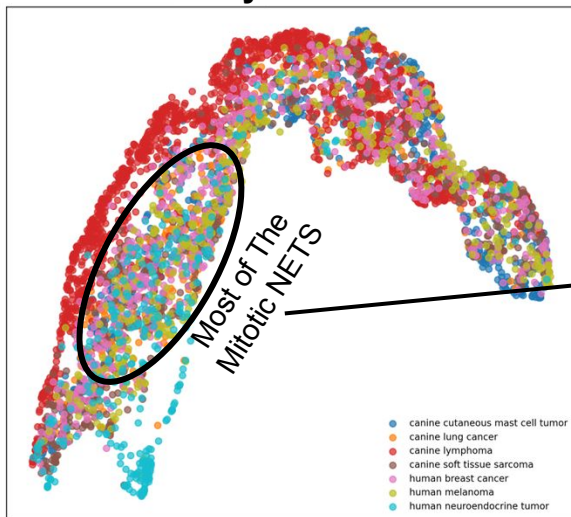
F1 Score per Tumor Type (Test Set)



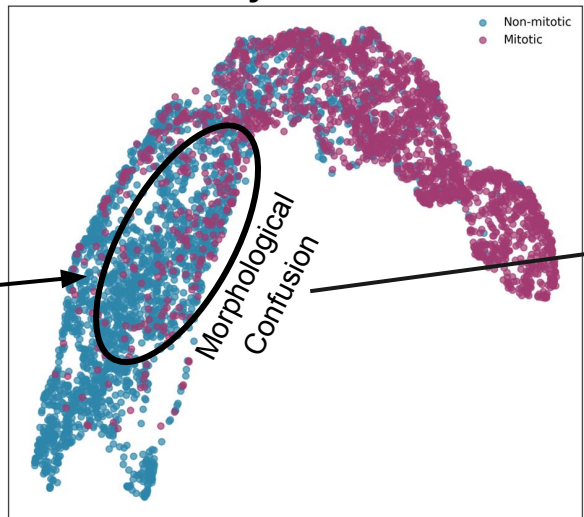
- **AUC (0.916):** high diagnostic discriminative power
- **Domain Generalization:** scanner, lab, and species noise successfully eliminated
- **The NET Gap:** Performance drop reflects unseen morphology and potential over-regularization

Failure Analysis & Takeaway

By Tumor



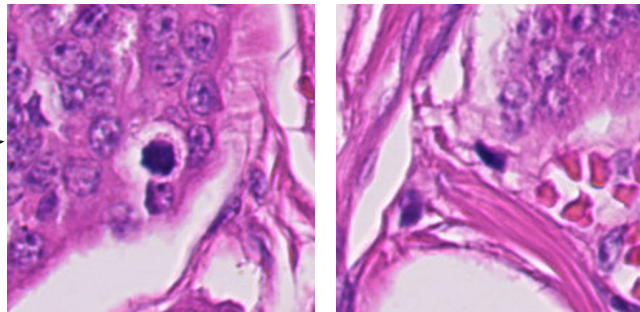
By Class



Dense, Hyperchromatic Nuclei

Mitotic NET

Non-Mitotic NET



Even *expert pathologists* disagree on NET grading up to 44% of the time

Takeaway

Multi-scale adversarial training achieves 0.916 AUC and eliminates scanner and lab noise. What remains is genuine morphological ambiguity that challenges even expert pathologists.



Thank you!

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